

HAEMIN PARK

Industrial Engineering and Management Science
Northwestern University

Developing **efficient** machine learning methods through optimization theory and algorithm design, with a focus on low-rank adaptation, federated and continual learning, and diffusion models.

Education

Northwestern University

Ph.D. in Industrial Engineering and Management Sciences | Expected June 2027

September 2022 - present | GPA 3.88/4.00

Seoul National University

Bachelor of Engineering | Double major in Industrial Engineering and Mathematics

March 2016 - August 2022 | GPA 3.87/4.30

Seoul Science High School

March 2013 - Feb 2016 | GPA 4.22/4.30

Selected Research

FedLoRU | Communication-Efficient Federated Low-Rank Update Algorithm and its Connection to Implicit Regularization

Transactions on Machine Learning Research (TMLR) | Accepted

2023 - 2025

- Analyzed low-rank nature of federated learning theoretically, and proposed parameter-efficient federated learning algorithm using low-rank updates.
- Designed FedLoRU to train low-rank client updates and periodically accumulate them at the server, allowing model adaptation to extend beyond a fixed low-rank subspace.
- Demonstrated that FedLoRU outperforms full-rank federated updates in settings with many clients, with its accuracy advantage widening as the number of clients increases across evaluated configurations, while reducing communication costs.

FedCGNM | Class-Grouped Normalized Momentum and Faster Hyperparameter Exploration

ICML 2026

2024 - 2025

- Developed class-grouped normalized momentum to reduce directional noise and mitigate class imbalance in federated optimization. Demonstrated that FedCGNM improves training in highly imbalanced training.
- Designed a hierarchical bandit method for client-specific resampling rates, enabling adaptive hyperparameter exploration under heterogeneous class distributions.

Fed-BRDECS | Privacy-Preserving and Heterogeneity-Aware Federated Deep Embedded Clustering

Preprint

2025 - 2026

- Developed a federated clustering method combining sample stability, balanced local sampling, and centroid restarting to address heterogeneous client data and aggregation noise.
- Studied cluster-composition leakage from shared statistics and model updates, and evaluated clustering for time-series anomaly detection.

Ongoing Research

Quantization of Few-Step Flow-Matching Models

Work in progress

2026 - present

- Characterized quantization-induced sampling trajectory drift in few-step flow models, examining how weight reconstruction error relates to errors in the learned velocity field.
- Developing a dual low-rank framework that reshapes weights before quantization and learns separate corrections for residual flow errors, with 3-bit and 4-bit experiments.

Continual Learning with Decoupled LoRA

Work in progress

2025 - present

- Investigating how separating low-rank update magnitude and direction affects adaptation and forgetting, with a focus on the relationship between parameter displacement and stability.
- Developed Decoupled LoRA with SAM (Sharpness-Aware Minimization) for continual learning by separating update magnitude and direction, improving adaptation to new tasks while reducing catastrophic forgetting through flatter and more stable low-rank update geometry.

Industrial Experience

Intel Corporation | AI Research Engineering Intern

Diffusion for generating rare defect images

June - September 2026

- Researched diffusion-based synthetic defect generation for industrial manufacturing images, addressing the scarcity of labeled anomaly samples and rare geometric pattern of chip manufacturing images.
- Built a diffusion pipeline for generating semiconductor defect images and segmentation masks from 10–50 labeled defects per class, using LoRA adaptation and lightweight spatial adapters. Extended coupled-generation approach with pseudo-normal reconstruction, normal-image conditioning, and latent blending to preserve chip geometry.
- Improved detection on held-out real images through filtered synthetic augmentation, increasing the performance about 18%.

Intel Corporation | AI Research Engineering Intern

Efficient vision models

June - September 2025

- Developed a Vision Transformer evaluation pipeline to compare inference latency, energy efficiency, and robustness to class imbalance for manufacturing inspection.
- Proposed adaptive model selection criteria for practical deployment in manufacturing-quality visual inspection.

Technical Skills

ML/AI Low-rank adaptation, federated learning, continual learning, model quantization, diffusion and flow matching, anomaly detection

Methods Optimization, matrix methods, parameter-efficient fine-tuning, adaptive hyperparameter exploration

Tools Python, PyTorch, Hugging Face Diffusers, Git, Docker